The effect of drought on the hazard into spousal violence

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Abstract

This paper investigates how droughts influence the onset of intimate-partner violence. We combine high-resolution precipitation data with duration data derived from household domestic violence surveys in India to estimate how negative income shocks impact the risk of marital violence. We find that a negative precipitation shock of a magnitude expected once every 4–5 years *decreases* the probability of first-time spousal violence by 0.037 percentage points in the formative years of marriage, representing a 70% reduction from the unconditional baseline hazard. As a complementary analysis, we revisit a recent study linking similar drought shocks to child marriage, applying updated methods and improved data. We now find that droughts reduce the hazard into early marriage both in Indian contexts practicing dowry payments and in Sub-Saharan African settings with bride-price customs. These effects are also largest for marriages occurring later in life. Our findings suggest a much more consistent relationship between drought and the timing of marriage across cultural contexts and that child marriage is likely more culturally coerced than economically motivated.

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"Human beings were creatures of habit and it was amazing the kind of things they could get used to."

Arundhati Roy, *The God of Small Things*

1 Introduction

The World Health Organization estimates that 27% of women worldwide will experience physical or sexual violence by an intimate partner in their lifetime, making it more prevalent than other public health priorities such as diabetes, depression, or breast cancer (Miller and McCaw 2019). Including expressions of emotional abuse, decades of research have established associations between intimate-partner violence (IPV) with severe medical and psychiatric conditions (Sugg 2015). Accordingly, the elimination of gendered violence and harmful gendered practices was identified as a priority target in the United Nations' 2030 Agenda for Sustainable Development with the incidence of intimate-partner violence and child marriage specifically identified as the measure of progress towards this goal (United Nations 2015).

A developing interdisciplinary literature has drawn attention to "economic abuse"—in which a person's economic dependence on a partner is instrumentalized against them—as a manifestation of and contributor to the prevalence of IPV (Stylianou 2018). For example, Adams et al. (2024) find that women's labor market outcomes precipitously diminish immediately upon cohabiting with an eventually-abusive partner. The authors interpret this result through a model in which the abuser intentionally creates economic dependence to disincentivize exiting the relationship. Aizer (2010) provides evidence that pay parity reduces physical violence against women.

This paper explores how shocks to this economic dependence relate to the occurrence of violence in a previously never-violent relationship. Here, I combine high-resolution weather data with comprehensive duration data on the incidence of intimate-partner violence to estimate how drought-induced negative income shocks may affect these health outcomes. I also replicate a recently published study relating these weather shocks to the risk of child marriage using updated methods and improved data.

Rural India presents an ideal first setting in which to study the malleability of gendered cultural norms to economic incentives. In addition to the wide availability of research products and infrastructure, this focus allows us to hone in on a region where gender inequities have proven remarkably resilient and where credit constraints and agricultural dependencies are well known to exacerbate the effects of economic shocks. The latter points are vital for our approach as these vulnerabilities combined with the sheer size and diversity of the country's people, cultures, and geography enables us to follow in an established tradition of using random drought shocks in developing countries as natural sources of measurable and localized identifying variation (Burke et al. 2014; Corno et al. 2020). In addition, a country marked by variation in gendered and transactional cultural practices such as dowry payment present opportunities to investigate channels through which gendered effects may operate. Finally, the common cultural practices of patrilocality—in which married couples cohabit with the groom's family—and multigeneration households present opportunities for follow-up studies on the influence of socialization and kinship structures.

My findings suggest that negative income shocks substantially reduce the hazard into marital violence. Specifically, a negative precipitation shock of a magnitude one might ordinarily expect

once every 4-5 years on average reduces the risk of a never-violent relationship becoming violent by 0.037 percentage points, representing a 70% reduction in the baseline hazard of this transition during the formative years of a marriage.

Updating the main result of Corno et al. (2020), I find evidence corroborating the original result that the same weather shocks reduce the hazard into child marriage among girls in India. But whereas the original study had reported opposite-signed effects in Sub-Saharan Africa, I find that these shocks also reduce the hazard into child marriage in Sub-Saharan Africa. I provide complementary results suggesting the practice of child marriage may be less motivated by environmental and economic factors than previously suggested, exemplifying the persistence of patriarchal norms despite economic development.

1.1 Related literature

An active area of research intersecting the economics of gender and development interrogates why substantial differences in gendered outcomes cannot be satisfyingly explained by their level of development. Though plenty of evidence corroborates the intuition that economic growth and conventional indicators of women's empowerment should co-move and positively impact one another, surveys of the literature on this gendered variety of modernization theory have concluded that "the interrelationships are probably too weak to be self-sustaining." (Duflo 2012). Prior work has shown this disconnect between economic and gendered progress is reproduced within countries: the correlation between per capita income and many of these measures are weak and inconsistent in sign across Indian states (Jayachandran 2021). Indeed, much academic attention has rightly concentrated on puzzles such as the labor-force participation rate of women in South Asia, which is markedly lower than in regions of comparable or lower levels of development and have remained so despite decades of sustained economic growth (Blackman et al. 2022).

In its many psychological and physical manifestations, the pervasiveness of IPV is well understood an acute measure of patriarchal culture yet it still presents additional complexities. For one, it does not neatly co-move with economic development: recent surveys by YouGov and the anti-IPV organization Frontline100 reported levels of tolerance for and experiences of IPV in high-income countries of the EU at comparable levels to those self-reported by respondents in rural India, our setting of focus. Nor does IPV incidence necessarily co-move intuitively with other conventional indicators of gender equality. Separate studies have claimed that otherwise culturally conservative countries and subnational regions within countries may report lower IPV incidence if they have proportionally higher Muslim populations; they ascribe the result to Islam's cultural aversion to alcohol (Kovacs 2018).

Two studies suggest that early-life exposure to differing family structures lead to measurable changes in gender norms. Fernández et al. (2004) finds that war-induced shocks to female labor-force participation, by normalizing working mothers, led male children of such working mothers to seek out working partners in their adulthood. Unterhofer and Wrohlich (2017) find that the takeup of paternity leave induce positive effects on attitudes towards gender norms both in the offspring of the fathers and to the untreated network of these fathers.

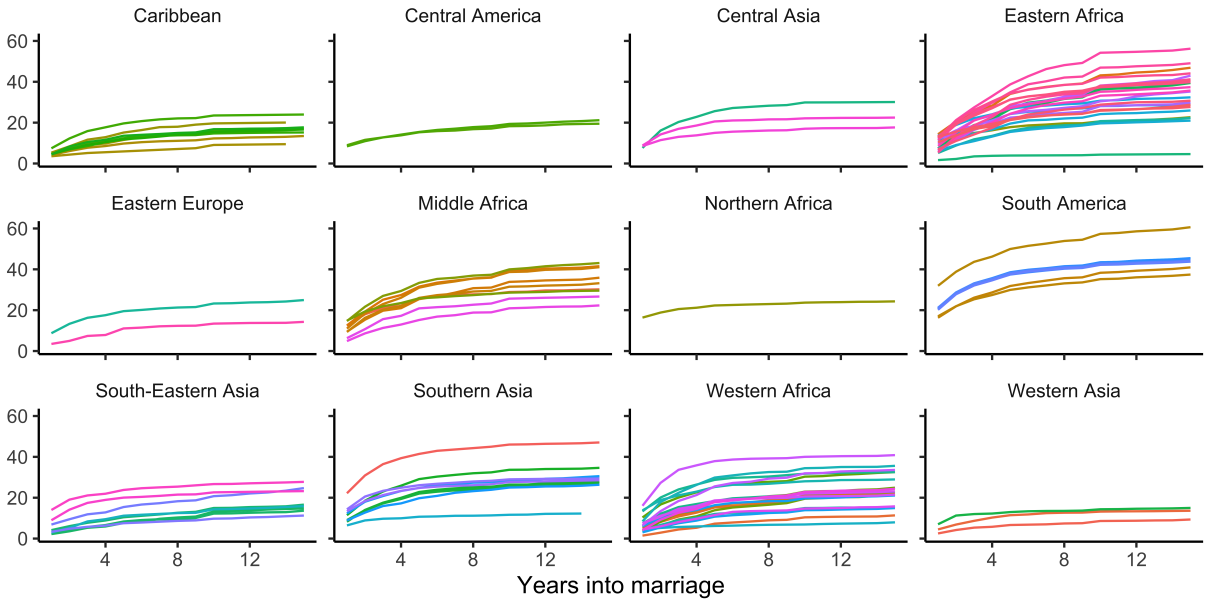
2 Data

2.1 Weather

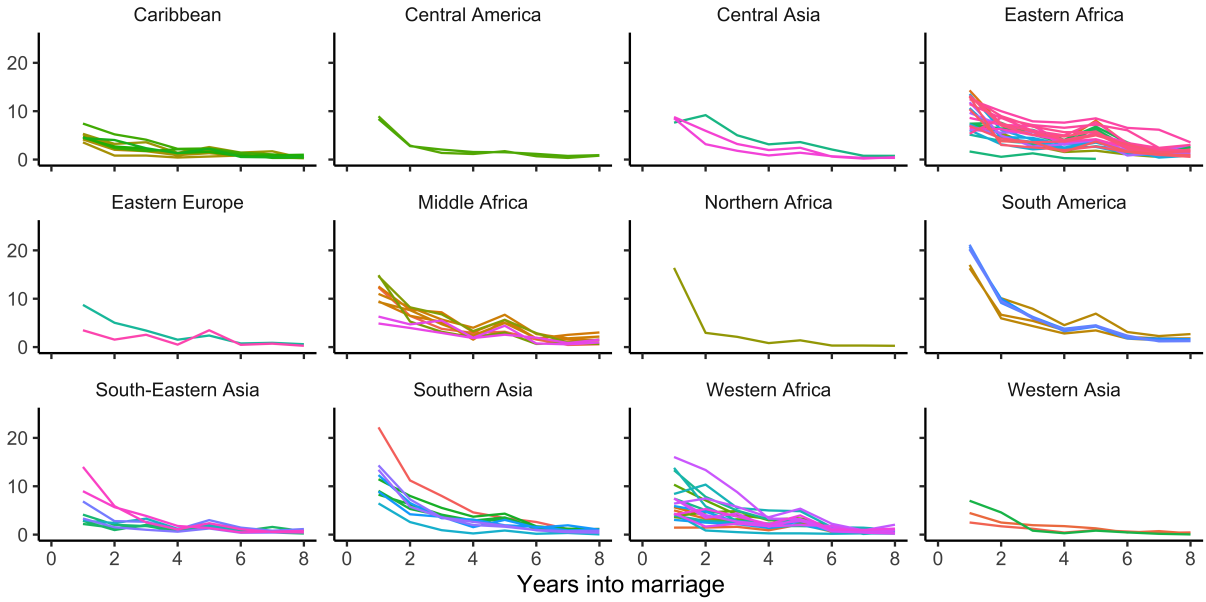
Precipitation data comes from CHIRPS (Funk et al. 2015), a $0.05° \times 0.05°$ gridded product spanning spanning the latitudes from 50°S to 50°N. Derived from a combination of infrared satel-

Figure 1: Incidence of self-reported domestic violence

(a) Cumulative percentage of marriages reporting violence



(b) Annual hazard into first-time violence



lite sensors, rain gauge data, and climate reanalysis models, CHIRPS is optimized for monitoring drought conditions in data-scarce regions; this fits our purposes well given our need to identify and differentiate between the historical precipitation experiences of thousands of rural locations (see Figure A1).

We make use of this extensive history in constructing the weather shocks which serve as the explanatory variable of interest in all our analyses. Because a given level of precipitation would be differently remarkable across the extremely diverse terrains of sub-Saharan Africa and the Indian subcontinent, an absolute level of precipitation may constitute flooding in one geography but relative dryness in another. For this reason, we follow Burke et al. (2014) , Corno et al. (2020) , and McGavock and Novak (2023) among others in defining 'droughts' relatively: for a given set of coordinates, we extract nearby precipitation observations from every month from 1981 onwards to fit a gamma distribution specific to the location. Then for any event of interest which we can locate in time and space, we can relate the weather observed in the preceding months or years to this local distribution and assign it a locally defined precipitation quantile.

In our main specifications, we use the 15th percentile in the local rainfall distribution as the threshold for qualifying as a relative 'drought'. While this construction is crude and arbitrary in some respects, it has the advantage of inducing similarly frequent shocks at all locations and can be easily scaled to be made more stringent or less restrictive by using different threshold percentiles to define the shock. We will demonstrate the sensitivity of our results to these alternative thresholds, effectively trading off magnitude of impact for sample size.

2.2 Domestic violence

The Demographic and Health Surveys (DHS) are implemented as a series of nationally representative cross-sections. Typically, all women between the ages of 15 and 49 years are selected from each household visited to undergo a detailed interview on topics relevant to public health and impact evaluation categorized into modules. Responses are anonymized but jittered GPS coordinates are provided at the village level; we use these to assign histories of weather exposure to a given respondent identifier relative to available dates such as the month-year of interview, birth, or cohabitation.

DHS surveys have now been conducted in more than 90 low- and middle-income countries but the waves administered in India are by far the most comprehensive with the number of respondents in a given round comparable to those in all of Sub-Saharan Africa; Figure A1 plots more than 28,000 unique cluster locations provided in its 2016 wave.

While each generation of surveys adheres to a standard format to enable comparisons across settings, some implementations will differ in their sampling methodology or incorporate additional modules on other public health topics. Of particular interest to the present analysis is the development and refinement of an optional domestic violence module, which is administered to one selected woman per household. Inspired by the Conflict Tactics Scale (Straus 1979; Straus et al. 1996), these respondents are asked whether they have been subject to specific acts of violence (e.g., kicking, striking with a weapon) and if so, their relationship to the perpetrator, if any. The incidence of domestic violence is quantitatively operationalized through discrete affirmative or negative responses to these questions (see Table A1 for summary statistics for variables relevant to domestic violence).

Both the appeal and limitation of this measure is in its simplicity. Tabulating binary answers to unambiguous questions is plausibly less susceptible to social desirability bias than earlier alternatives but also removes contextual information and subjectivities that may meaningfully qualify a notoriously complex social phenomenon (Smith, Tessaro, et al. 1996; Smith, Earp, et al. 1995).

Results of this data collection cannot differentiate between chronic battery nor does it consider the meanings and dangers which victims assign to these events.

To investigate the hazard into domestic violence, we make particular use of a single uniquely informative question[2]. After being asked about experiences with a variety of physically violent acts, respondents who answered any question affirmatively were then asked in which specific year of their marriage they first experienced any of these acts. Figure A2 depicts the distribution of answers to this question for hundreds of survey waves in 12 global regions. Other than mild evidence of bunching at marriage-years divisible by 5, the distributions are consistent across the first nine years and remarkably well-distributed exhibiting approximately exponential decreases from a peak in the first year of marriage. Figure 1a depicts the implied cumulative distribution of marital violence over time and Figure 1b charts the implied annual hazard into first-time violence: conditional on never previously having been violent, the probability of a marriage becoming violent for the first time is clearly largest in the early years of a marriage. For this reason, we specify the first four years as the baseline risk period for our analysis.

By placing this potentially critical event in time, we can infer more about its context by relating it to the timing of other events. For example, respondent and family-member month-years give us information on the ages of the respondent, perpetrator, and household kinship structure at the time of the event. The sample of respondents for our main results come from the two India survey waves which both provide GPS coordinates and which implemented a version of the domestic-violence module which included this now-defunct timing variable.

2.3 Early marriage

We also draw from the DHS surveys to make inferences about the effect of these shocks on the timing of marriage, updating the main result of Corno et al. (2020) using much more recent and higher-quality data.

The original analysis relied on the 1998-1999 survey wave for Indian respondents and 1994-2013 data for Sub-Saharan African countries. In comparison, we use Indian data from the 2015-16 and 2019-21 waves and all available sub-Saharan African survey waves for which GPS coordinates were available. Our estimation sample constitutes 6.8-8.2 million respondent-age observations for India and 5.1-6.3 million for sub-Saharan Africa compared to 329,586 and 1,799,037 for the original analysis. In addition, this older generation of Indian DHS surveys purposely only interviewed women who were married at the time of the interview, which would strictly overstate estimates of the annual hazard into marriage.

Critically, this earlier vintage of Indian DHS data also did not come with GPS coordinates, meaning respondents could only be geographically matched to highly aggregated districts roughly 5000 square kilometers in area on average. In contrast, GPS coordinates provided in later waves are guaranteed to lie within 10 kilometers of the respondent's true location. The higher quality weather data described above and these much more precise DHS locations mean that our shocks are far more localized to the true locations of the respondents. For additional context, we plot baseline annual hazard rates and cumulative rates of marriage in Figure A3.

[2]Versions of this variable which appeared in the earliest domestic violence surveys restricted focus to sexual violence before it was broadened to consider any of the physical acts enumerated in Table A1. This question has now been discontinued, having been phased out of DHS survey questionnaires beginning with the launch of DHS-8 in 2018.

3 Methods

Our main analysis is an empirical test of whether drought shocks impact the probability of a marriage becoming violent for the first time. While our data is sourced from the repeated cross-sections of the DHS program, the required duration data can be constructed by exploiting the dynamic information implied by the timing variable highlighted in the previous section.

Specifically, we construct a pseudo-panel where each observation corresponds to the experiences of a respondent i by their tth wedding anniversary. With the provided household GPS coordinates, we can then produce a timeline of drought exposure specific to each respondent under the assumption that the respondent did not move to a substantially different geography over this period. Our main explanatory variable, $D_{i\ell t}$ is a binary indicator for whether the volume of precipitation which occurred in the period t since respondent i was married constituted a relative drought in the history of rainfall specific to location ℓ.

We estimate Equation (1) to estimate the effect of drought exposure on the incidence into first-time domestic violence, extending duration data methods employed in other studies of hazards into gendered outcomes such as child mortality (Currie and Neidell 2005), child marriage (Corno et al. 2020), and female genital cutting (McGavock and Novak 2023).

$$V_{it} = \sum_{p=0}^{P} \beta_p D_{i,\ell,t-p} + \boldsymbol{\gamma}^\top \mathbf{W}_{i,\ell,t} + \lambda_i + \phi_\ell + \eta_t + \varepsilon_{i,\ell,t} \tag{1}$$

The outcome variable V_{it} is a binary indicator of whether respondent i had experienced marital violence for the first time in the 12 months preceding their tth wedding anniversary. All respondents are observed in our dataset for the first year of their marriage $t = 1$. Respondents who experienced domestic violence in this period 'drop out' from our dataset since they are definitionally no longer at risk of first-time violence. We repeat this process for each $t > 0$ until all respondents have experienced violence or up to a horizon $\bar{t}$ defining the end of the risk period of interest which we had determined from Figure 1b to be the fourth year of marriage in our main specification though we also present results under alternative values of $\bar{t}$.

Our main coefficients of interest are the set $\{\beta_p\}_{p=0}^{P}$, corresponding to the absolute change in the annual hazard into domestic violence causally attributed to a drought shock and its P lags. These coefficients can then be divided by the unconditional hazard rates depicted in Figure 1b over the risk period of analysis to determine the proportional change in the hazard rate attributable to the shocks.

In our main specification, the matrix $\mathbf{W}_{i,\ell,t}$ of controls includes indicators for the survey wave the respondent data was collected for, the panel marriage-year t the observation pertains to, the village-level 'cluster' whose coordinates are assigned to the respondent in the data, and the calendar year in which the respondent was married. Errors are clustered at the level of the GPS coordinates from which we derive the treatment variable. In the Indian data, a given cluster's coordinates are assigned to 17 different respondent households on average[3]. Given that our drought shock treatment is assigned on the basis of these coordinates and likely highly autocorrelated across space, households assigned the same cluster are likely to exhibit systematically more correlated responses. We also estimate the equivalent of Equation (1) using entry into child or early marriage as the outcome and women's adolescent years from ages 12-18 as the risk period of analysis. We also include lagged shocks to incorporate dynamic effects not considered in the original anal-

[3]While this would ordinarily correspond to hundreds of female respondents, the domestic violence module of the DHS is unique in only being administered to one randomly selected woman in each household.

ysis by Corno et al.

4 Results

4.1 Hazard into intimate-partner violence

Our main results are summarized in Section 4.1, representing the effect of a drought on the annual probability of a non-violent marriage becoming violent. Estimated effects across specifications are significantly negative across the different sets of fixed effects considered, suggesting that the experience of a drought in the early years of marriage actually reduces the hazard into marital violence.

Estimates are most pronounced for the first lag of the drought shock, implying that new wives are approximately 0.04 percentage points less likely to experience first-time violence with the occurrence of a drought in the first four years of marriage. For our preferred specification, Model (3), this estimate is 0.037 [0.016, 0.058]. Compared to an unconditional hazard over this period of 0.053 representing, this represents a substantial 70% reduction in the hazard into marital violence.

Figure 2 depicts the sensitivity of these estimates to different choices about the relevant risk period and threshold for defining a drought with our preferred specification highlighted in blue and error bands corresponding to a 95% confidence interval. In Figure 2a, defining the risk period over a sufficiently long period nullifies the drought effect suggesting that environmental conditions (and plausibly the income channels they operate through) can influence the propensity for a marriage only in its earliest formative years. One interpretation of this pattern is that for marriages which have remained non-violent for sufficiently long, breaks from this established norm are more difficult to influence.

Figure 2b shows that, unsurprisingly, the less stringent the definition of a drought (i.e., as the threshold percentile increases), the smaller the estimated effect it has on the hazard into marital violence. On the other hand, for sufficiently stringent definitions of drought, the frequency of the 'treatment' variable diminishes, reducing the precision of these estimates. So while years with precipitation less than the 5th percentile in the local historical distribution lead to larger point estimates, the 95% confidence interval ranges from an 8.5% reduction in the hazard to its complete elimination.

Table A2 summarizes results varying the number of lags of the drought variable included and estimating effects separately for the two waves of the Indian DHS survey. We find that regressions which omit the lagged shocks return null effects while including additional lags does not provide additional explanatory power nor does it affect the estimate on the first lag of the drought. The drought effect, then, is entirely transitory and operates with a one-period lag, similar to the drought effect on democratic transition documented by Brückner and Ciccone (2011).

4.2 Hazard into early marriage

Section 4.2 reports the results of an analogous exercise, measuring the effect of droughts on the hazard into child marriage for women between the ages of 12-18. This represents an update and extension of the main results from Corno et al. (2020) on much newer and improved data[4].

We report new strong evidence that droughts reduce the hazard into child marriage in India, supporting the original finding of Corno et al. Because of differences in data collection methods—

[4]The original analysis used India's 1998-99 DHS survey data compared to our use of India's 2015-16 and 2019-21 rounds. We make note of the substantive differences in data quality between these rounds in Section 2.

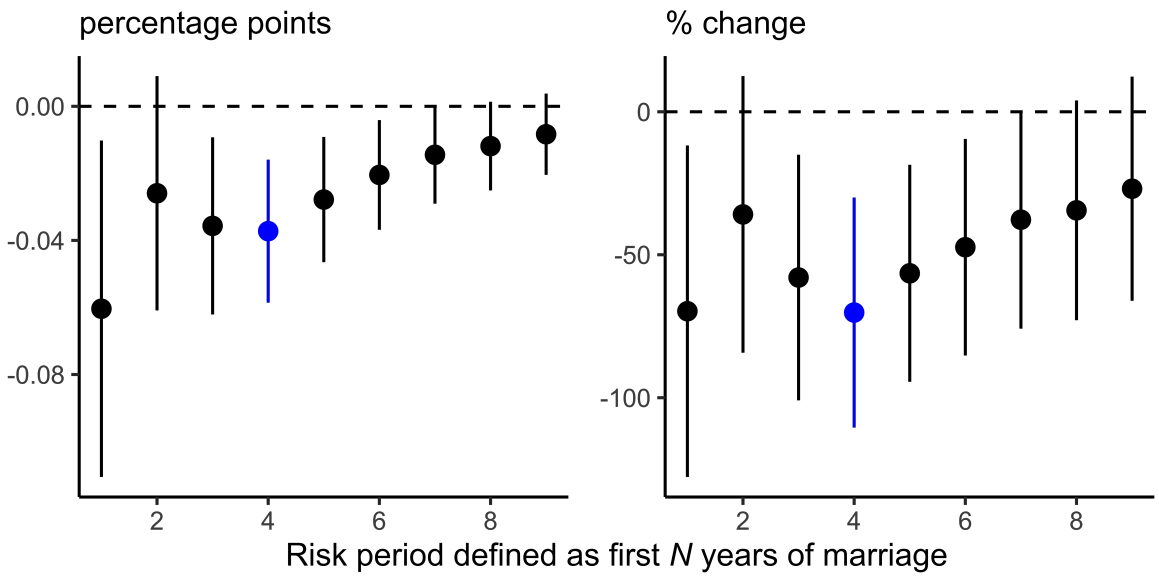
Table 1: Effect of drought on the hazard into marital violence

Dep. variable:	Violent			
Model:	(1)	(2)	(3)	(4)
Variables				
Drought	-0.0004***	-0.0002*	-0.0001	-0.0001
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Drought, L1	-0.0006***	-0.0004***	-0.0004***	-0.0004***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Drought, L2	-0.0002·	2.3×10^{-5}	-6.26×10^{-5}	-5.48×10^{-5}
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Fixed-effects				
Survey wave (2)	Yes	Yes	Yes	Yes
Panel anniversary (4)	Yes	Yes	Yes	Yes
Cluster (56,073)		Yes	Yes	Yes
Year married (38)			Yes	Yes
Age at marriage (51)			Yes	Yes
Region-Marital cohort (291)				Yes
Fit statistics				
Outcome mean	0.00538	0.00538	0.00538	0.00538
Observations	3,539,064	3,539,064	3,539,064	3,539,064
Adjusted R^2	0.00068	0.02577	0.02600	0.02607
Within R^2	1.85×10^{-5}	5.34×10^{-6}	3.93×10^{-6}	3.65×10^{-6}

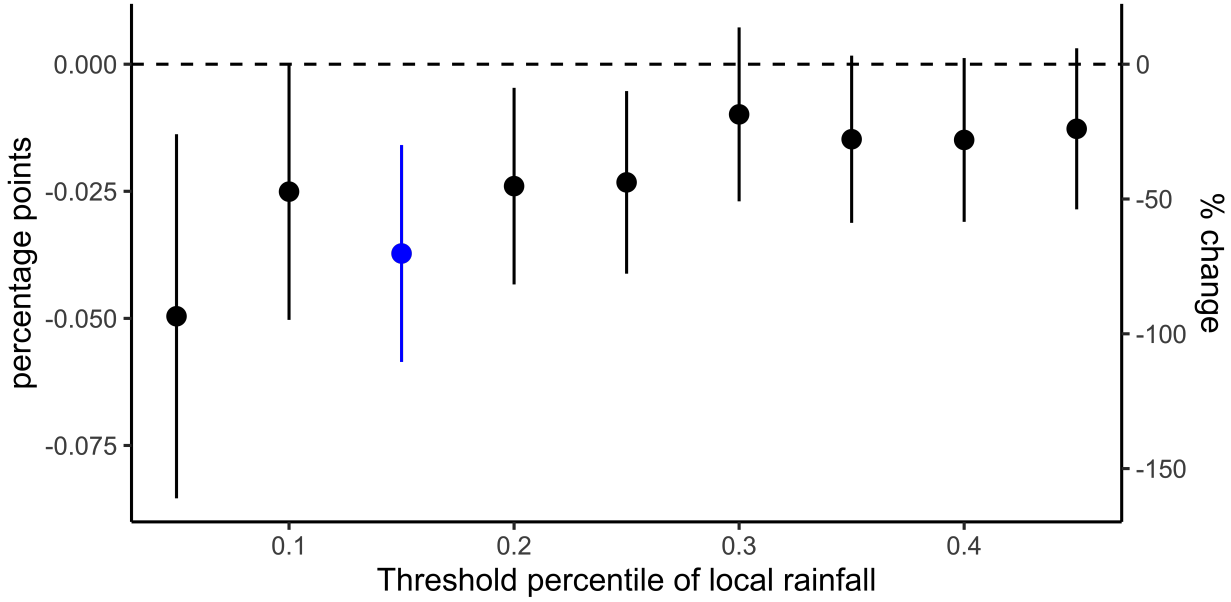
Clustered (Cluster) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, .: 0.1*

Figure 2: The effect of drought on the hazard into marital violence

(a) Varying reference period of risk



(b) Varying threshold intensity of drought



For all panels, estimates in blue correspond to our main specification which defines the period of risk as the first four years of marriage and defines a drought by whether the volume of precipitation in the preceding 12 months fell below the 15th percentile in the historical distribution local to the the respondent's cluster GPS coordinates. Vertical bars span a 95% confidence interval.

Table 2: Effect of drought on the hazard into child marriage

Dep. variable:	Married					
Region:	India			Sub-Saharan Africa		
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Drought	-0.0011***	-0.0009***	-0.0012***	-0.0012***	-0.0015***	-0.0011**
	(0.0002)	(0.0002)	(0.0003)	(0.0003)	(0.0003)	(0.0003)
Drought, L1		-0.0010***	-0.0018***		-0.0025***	-0.0025***
		(0.0002)	(0.0003)		(0.0003)	(0.0003)
Drought, L2			0.0009***			-0.0022***
			(0.0003)			(0.0003)
Fixed-effects						
# Cluster	56,076	56,076	56,076	43,883	43,877	43,872
# Panel age	7	7	6	7	7	6
# Birth year	41	41	40	43	42	40
# Country	1	1	1	36	36	36
# Survey wave	2	2	2	106	106	106
Fit statistics						
Outcome mean	0.04919	0.04933	0.05724	0.05978	0.05966	0.06878
Observations	8,237,805	8,165,320	6,815,956	6,267,971	6,166,258	5,127,306
Adjusted R^2	0.05647	0.05662	0.05456	0.06722	0.06729	0.06635

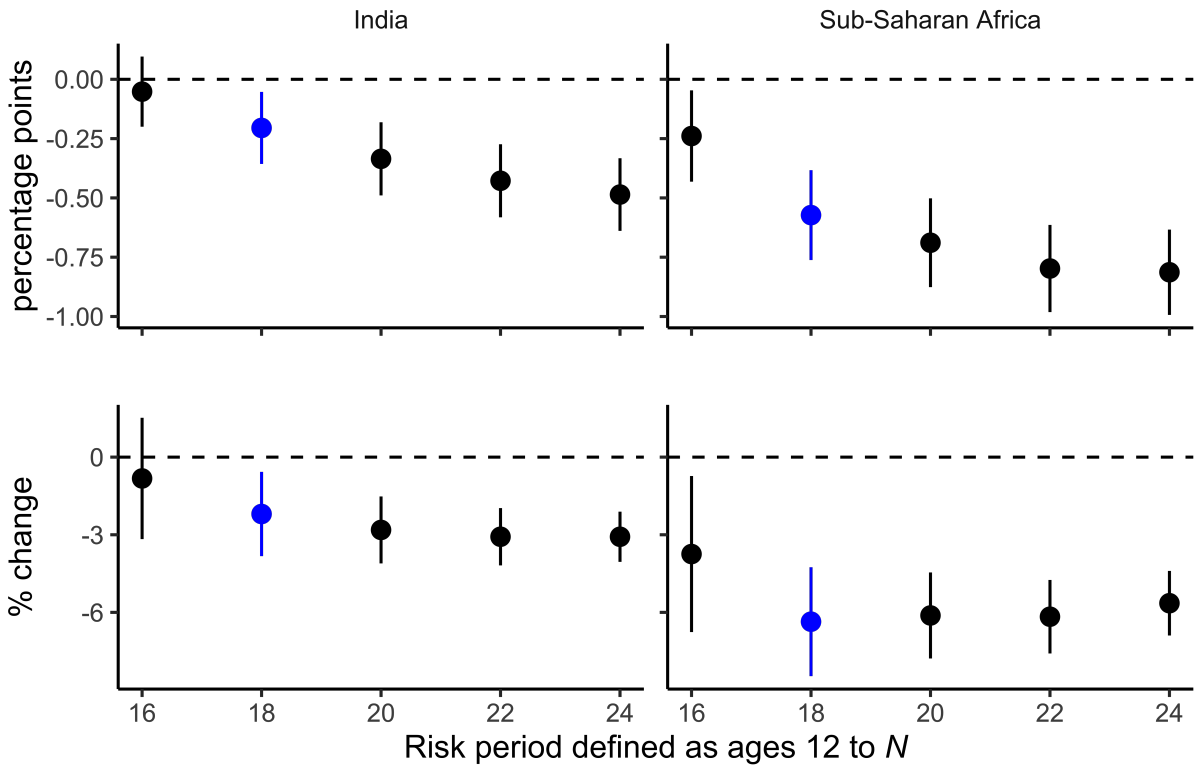
Clustered (Cluster) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, .: 0.1*

for example, the 1999 Indian DHS survey did not interview single women—the older analysis will generally imply larger conditional and unconditional hazard rates. We also incorporate dynamic effects of lagged droughts which were not considered in the original analysis.

Despite these differences, recovered effects in India are on the order of the original results though significantly reduced: in our preferred specification (Column 3), the cumulative effect of droughts experienced between the ages of 12-18 reduces the hazard into child marriage among young women in India by 0.21 percentage points, representing a 2.19% reduction from an unconditional hazard rate of 9.3%. In the original analysis, droughts were found to have reduced this hazard by 0.46-0.47 percentage points, a 4.2-4.3% decline relative to a hazard rate of 10.9 percentage points.

We are unable to reproduce the finding of opposite-signed effects in Sub-Saharan Africa, the main result of the original analysis which attributed these opposite dynamics to consumption smoothing induced by opposite customs of marriage payment. Instead, we find that the hazard into child marriage is even more substantially reduced in the sub-Saharan African sample if anything: a 0.57 percentage-point reduction from a baseline hazard of 9.0%, representing a reduction of 6.4%.

Figure 3: The effect of drought on the hazard into early marriage



For all panels, estimates in blue correspond to our main specification which defines the period of risk as ages 12-18. Estimates in black vary the upper bound of this risk period. Results in the top row multiply coefficient estimates by 100 for more intuitive interpretation in units of percentage points. Results in the bottom row divide these hazard impact estimates by the unconditional hazard into first-time violence. Vertical bars span a 95% confidence interval.

Figure 3 plots the sensitivity of these results to broader definitions of child marriage, finding that a drought event reduces this hazard even more at later ages.

5 Conclusion

This paper documents evidence that droughts during the formative years of a marriage reduce the annual hazard into domestic violence in India by 70%. Secondarily, we also find that the same shocks reduce the hazard into child marriage in India and Sub-Saharan Africa, suggesting these cultural practices may not be as motivated by the direction of marriage payments as suggested in recent literature.

These findings also recall the finding in McGavock and Novak (2023) that similarly constructed shocks marginally increase the propensity for families in Sub-Saharan Africa to forego the practice of female genital cutting. As the authors of that paper note, these results present something of a challenge in that they do not immediately prescribe policy responses to the social problems at subject; for example, it would be obviously absurd to use these results to advocate for acute reductions in income or to increase economic vulnerability to extreme weather events as a means of mitigating these harms.

Our sensitivity analyses suggest timing is likely to be an important consideration in reducing the incidence of these harmful traditions. The earliest years of a marriage appear to be the most formative for determining patterns of violent behavior. This would imply that efforts to discourage intimate-partner violence would be most effective concentrating on young marriages; after the first four years of marriage, the hazard into first-time violence is already substantially reduced as shown in Figure 1b and those that do take a violent turn are less likely to be a result of economic or environmental influence (Figure 2a).

For a number of high-population regions of the world, the hazard into child marriage is also high at early ages, as depicted by the density between ages 12-16 in Figure A3. However, Figure 3 shows this is also the age range where marriage rates are least influenced by external conditions, especially in India where we estimate a null effect of droughts. This would suggest then that the prospects for economic incentives as a means of curtailing or eventually eliminating this particular cultural practice are limited and alternative approaches should be prioritized.

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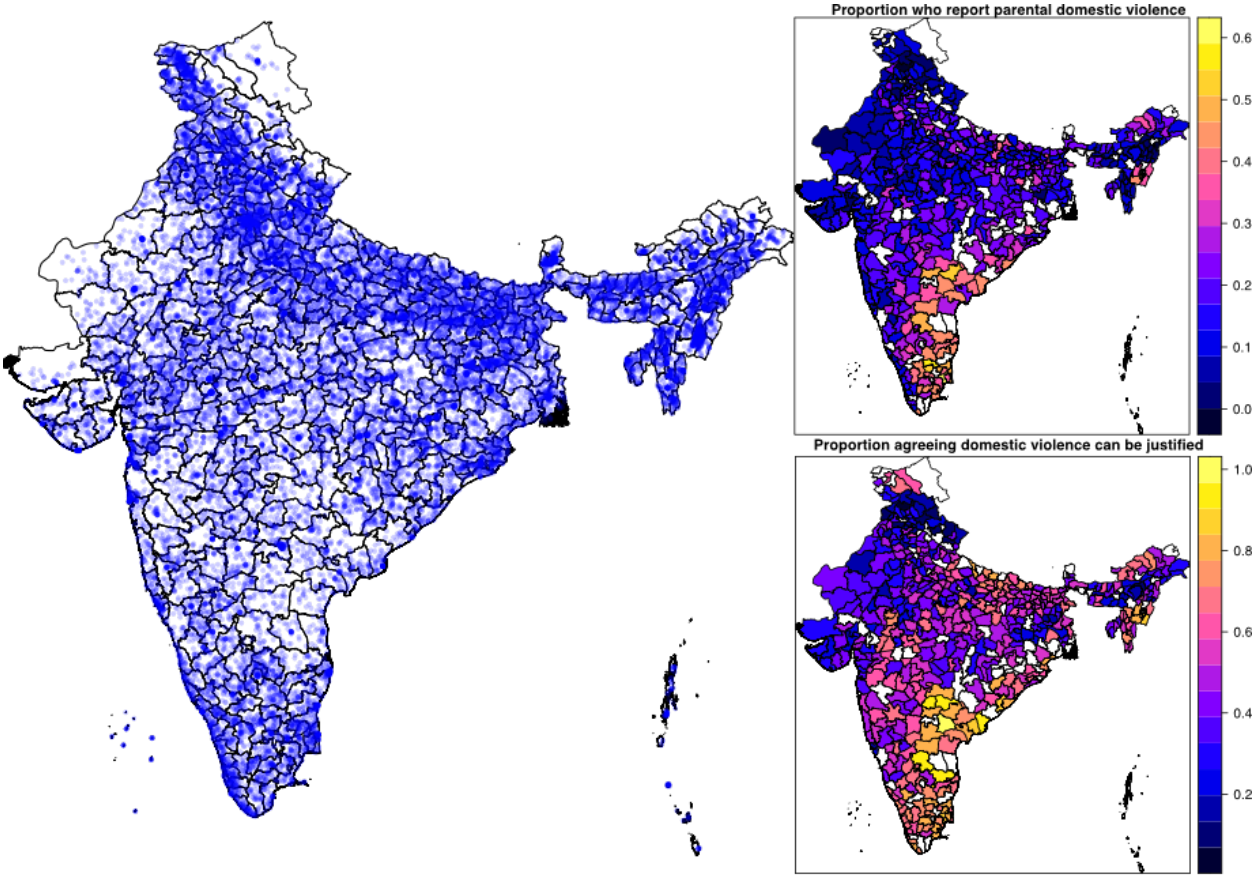
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A1 Data

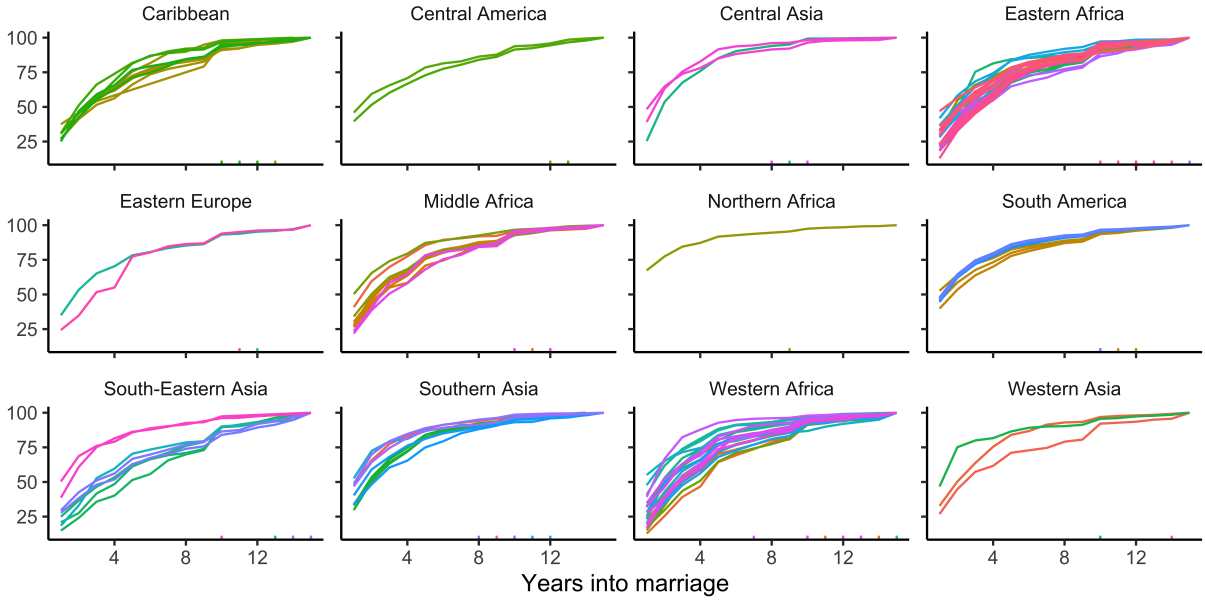
Figure A1: Spatial coverage of Indian DHS surveys



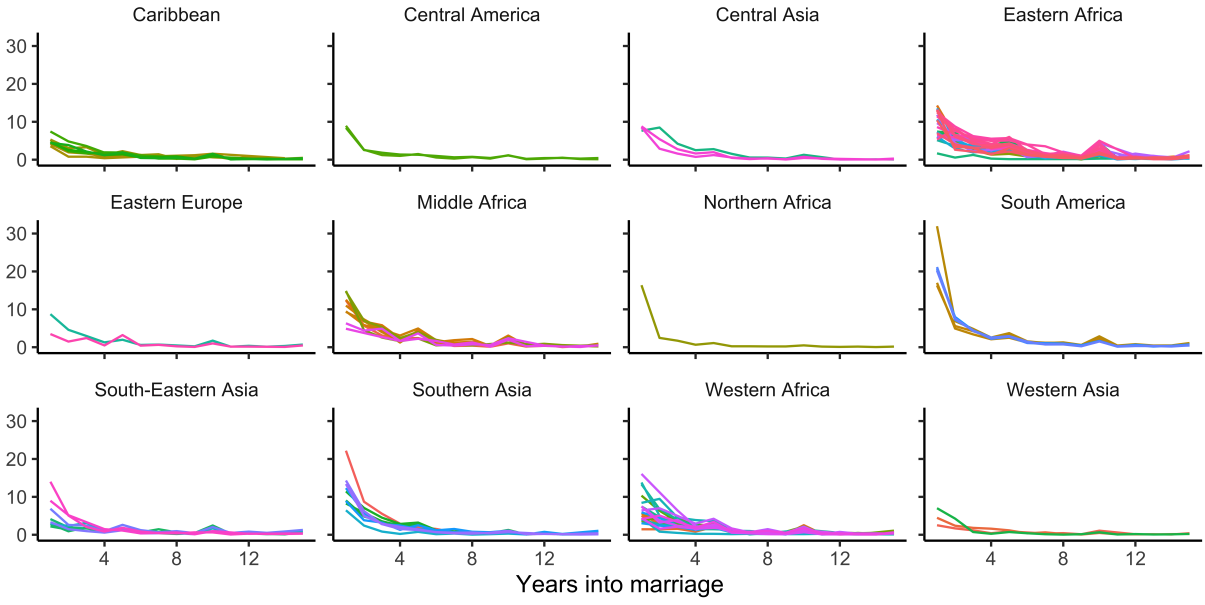
Map on left plots over 28,000 unique clusters from the 2015-16 wave of the Indian DHS surveys. Maps on right aggregates responses of women selected for the domestic violence survey module to the district level where available.

Figure A2: Incidence of marital violence, conditional on ever reporting violence

(a) Cumulative percentage of marriages violent



(b) Timing of first violent event



For all similar plots, colors distinguish countries within a global region, which may have multiple survey waves.

Table A1: Categorization of relevant questions from DHS domestic violence module

	Sub-Saharan Africa		India	
	Obs	Mean	Obs	Mean
IPV justified if wife:				
Argued with husband	116,599	0.692	86,520	0.719
Burnt food	116,713	0.826	86,678	0.813
Neglected child	116,891	0.657	86,786	0.670
Refused to have sex	116,341	0.716	86,049	0.860
Went out without permission	116,722	0.667	86,718	0.737
Wife justified in refusing sex if:				
Infidelity	97,264	0.357	85,594	0.152
Not in mood	28,271	0.361	85,176	0.175
Partner has an STI	33,861	0.204	84,387	0.141
No birth control	96,691	0.209	0	–
Recently gave birth	10,524	0.214	0	–
Household decisions wife has influence over				
Wife's money	51,924	0.879	25,969	0.855
Husband's money	87,411	0.463	80,850	0.769
Household purchases	101,126	0.556	81,876	0.755
Visits to family/friends	101,121	0.653	81,876	0.787
Contraceptive use	49,250	0.851	57,666	0.921
Health decisions	99,593	0.593	81,876	0.774
Sexual consent	84,654	0.351	0	–
Controlling behavior by husband				
Accuses wife of infidelity	113,060	0.228	86,973	0.085
Restricts contact, family	113,181	0.123	86,998	0.122
Restricts contract, friends	113,171	0.176	86,983	0.172
Exhibits jealousy	112,706	0.520	86,754	0.240
Access to money	41,389	0.209	86,898	0.201
Demands knowledge of whereabouts	113,164	0.405	86,923	0.166
Experience of physical violence, last 12 months				
Pulled hair / Twisted arm	99,827	0.077	87,163	0.129
Kicked	113,896	0.116	87,163	0.098
Punched	114,725	0.108	87,163	0.095
Pushed	115,554	0.129	87,163	0.139
Slapped	114,740	0.249	87,163	0.297
Strangled	115,545	0.035	87,163	0.021
Hit with a weapon	108,377	0.027	87,163	0.013
Experience of emotional violence, last 12 months				
Humiliated	114,306	0.173	87,163	0.107
Insulted	100,316	0.241	87,163	0.082
Threatened	114,288	0.102	87,163	0.057

A2 Supplementary results

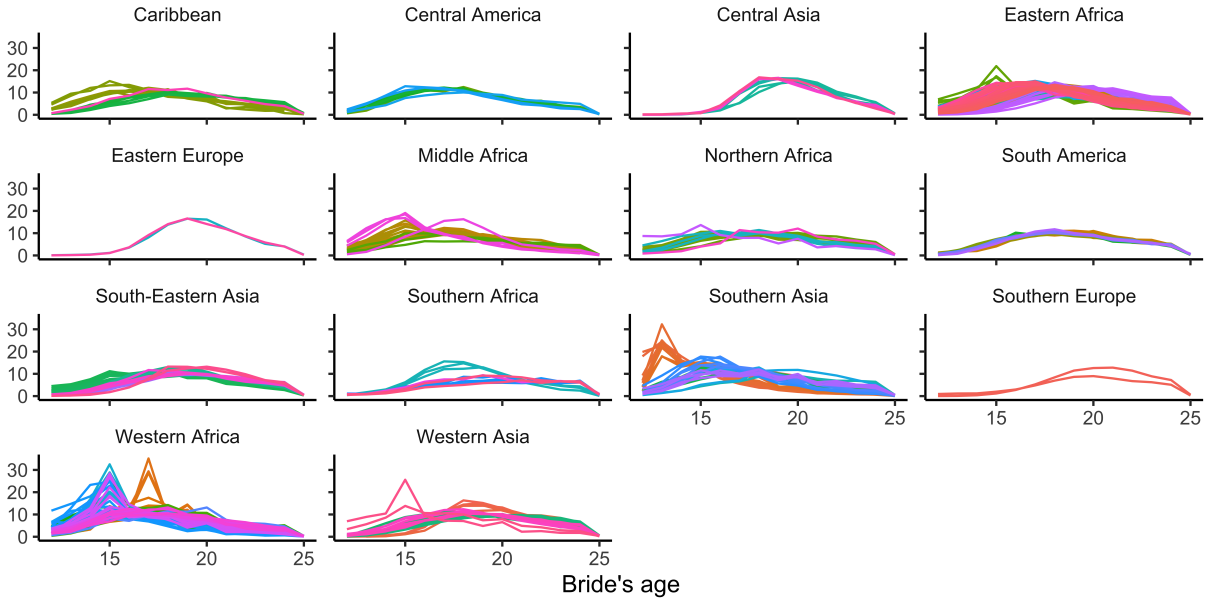
Table A2: IPV hazard results, cumulatively adding lags

Dep. Variable:		Violent	
Model:	(1)	(2)	(3)
Variables			
Drought	-0.0002	-0.0001	-0.0001
	(0.0001)	(0.0001)	(0.0001)
Drought, L1		-0.0004***	-0.0004***
		(0.0001)	(0.0001)
Drought, L2			-6.26×10^{-5}
			(0.0001)
Fixed-effects			
Survey wave (2)	Yes	Yes	Yes
Anniversary (4)	Yes	Yes	Yes
Cluster (56,073)	Yes	Yes	Yes
# Year married	40	39	38
# Age at marriage	52	52	51
Fit statistics			
Outcome mean	0.00538	0.00538	0.00538
Observations	3,550,939	3,546,383	3,539,064
Adjusted R^2	0.02592	0.02593	0.02600
Within R^2	5.38×10^{-7}	3.7×10^{-6}	3.93×10^{-6}

Clustered (Cluster) standard-errors in parentheses
*Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, .: 0.1*

Figure A3: Incidence of early marriage by age

(a) Annual hazard into early marriage



(b) Cumulative percentage of women married

